

Nazmul Alam Diptu

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RESEARCH SUMMARY

Focused on mathematically grounded, efficient transformer inference and long-context optimization.

RESEARCH PROJECTS

Mathematical Foundations of Efficient Transformers

- Developing a theoretical framework for transformer inference, emphasizing KV cache compression and attention scaling.
- Curating rigorous derivations in linear algebra and optimization to advance doctoral research on LLM efficiency.

EXPERIENCE

North South University

May 2019 – May 2025

Lab Instructor

Dhaka, Bangladesh

- Instructed 1000+ undergraduates in **core CS theory** and algorithmic problem-solving; led curriculum design and assessment strategies.

Me-Solshare Ltd

Nov. 2019 – Aug. 2021

Jr. Data Scientist

Dhaka, Bangladesh

- Engineered **high-accuracy CNNs** for time-series energy forecasting, utilizing rigorous statistical evaluation methods.

Power Ledger Pty Ltd

Oct. 2022 – Feb. 2024

Product Analyst

Remote

- Managed **technical data analysis and QA** for a blockchain platform, facilitating cross-functional alignment.

PUBLICATIONS

End-to-End OCR via Synthetic Dataset Generation (*IJCCI 2020, Springer*)

- Trained a **deep-learning OCR model** on synthetically generated noisy images; improved robustness in low-quality document recognition | [doi:10.1007/978-981-15-3607-6_41](https://doi.org/10.1007/978-981-15-3607-6_41)

Population Estimation of Rohingya Refugees via ANN (*VJCS 2019, World Scientific*)

- Modeled refugee population using **Artificial Neural Networks** on satellite and census data; applied to humanitarian resource planning | [doi:10.1142/S2196888819500246](https://doi.org/10.1142/S2196888819500246)

EDUCATION

North South University

2015 – 2018

B.Sc. in Computer Science and Engineering

- Honors:** CGPA 3.62/4.00. **Core:** ML, Algorithms, Databases, Systems, OOP.
- Certifications:** Deep Learning Specialization (DeepLearning.AI), Machine Learning Specialization (Coursera).

SKILLS

Languages : Python, C/C++, Java, JS/TS, SQL

ML & Research : PyTorch, TensorFlow, NumPy/Pandas, SciKit-Learn, MLflow

Tools & Cloud : FastAPI, Docker, PostgreSQL/MongoDB, AWS, Linux, $\LaTeX$